Ilyas Khan

AI/ML Engineer · Full-Stack Developer · Embedded Systems

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PROFILE

AI/ML engineer and full-stack developer with 34+ shipped projects spanning machine learning pipelines, NLP systems, autonomous embedded hardware, and production web applications. Built and deployed a multi-modal cancer classifier that achieved perfect AUC and won NeoHack 2025. Shipped a RAG-powered support agent for a real client. Implemented transformer self-attention from scratch, built Word2Vec and text-to-speech systems from first principles, and architected an open-source engine that converts natural language into working spreadsheet applications. Operates across the full stack — from Pixhawk flight controllers and servo actuators to FastAPI backends and React frontends.

TECHNICAL SKILLS

Languages	Python, C++ (OOP, DSA), TypeScript, JavaScript, SQL
AI / ML	scikit-learn, Deep Learning, LangChain, SHAP / Explainability, Word2Vec, TF-IDF, NLP Pipelines
NLP / Speech	Text-to-Speech (TTS), Word Embeddings, Transformer Self-Attention, Tokenization, Sequence Models
Full-Stack	React, FastAPI, Flutter, Supabase, Celery, Redis, REST APIs
Data / Vector	MySQL, Supabase, Pinecone (vector store), Pandas, Jupyter, EDA
Automation	n8n, Gemini API, Claude API, Workflow Orchestration, RAG Pipelines
Embedded	Pixhawk / ArduCopter, GPS Waypoint Navigation, Servo Control, F450 Quadcopter
Systems	Linux, Git, GitHub, Docker, Google Cloud Console, Blender 4.2 (rendering)

SELECTED PROJECTS

Multi-Modal Cancer Omics Classifier

🏆 NeoHack 2025 Winner

Python · scikit-learn · Pandas · Jupyter — 2025

Built a multi-modal ML pipeline fusing clinical, genomic, and transcriptomic data to predict cancer outcomes. Engineered feature extraction across three distinct data modalities, handled class imbalance, and tuned ensemble classifiers. Achieved a perfect AUC score on the evaluation set. Built under hackathon time constraints.

PlanCraft AI — Architecture Design Platform

SHIPPED

Flutter · FastAPI · Gemini 2.5 Flash · Celery · Redis · Blender 4.2 — 2025

Designed and built an AI-powered architecture platform that generates 2D and 3D home layouts from natural-language input. Architected the backend with FastAPI and an async Celery + Redis task queue to handle Blender render jobs. Built the Flutter mobile frontend and produced a full marketing deck.

SpreadsheetAppBuilder (SAB)

IN PROGRESS

Python · UNO API · Office.js · JSON Blueprint Pipeline — 2026

Architected an open-source engine that converts natural-language prompts into working spreadsheet applications. Designed a JSON blueprint pipeline that fully decouples the AI generation layer from the rendering layer. Ships with dual adapters: LibreOffice via Python/UNO and Excel via Office.js. Swap renderers without touching generation logic.

DocSmith — AI Document Automation

SHIPPED

Python · LLM Tooling · Prompt Engineering — 2025–2026

Built an AI-driven document generation system that produces structured professional documents from templates and data inputs. Engineered prompt templates for deterministic output quality, debugged configuration-level compliance defects, and shipped a fully audited codebase with 11 bug fixes across prompt templates and core modules.

Autonomous Parcel Delivery Drone (APDD) CAPSTONE

Pixhawk 2.4.8 · ArduCopter · F450 Frame · MG90S Servo — 2026

Engineered a GPS-guided quadcopter for autonomous last-mile parcel delivery. Configured the Pixhawk 2.4.8 flight controller with ArduCopter firmware for waypoint navigation. Designed and integrated a servo-actuated parcel release mechanism. Authored a full technical research report. Built on real hardware — not simulation.

AI Customer Support Agent — AJ Digital Agency CLIENT DEPLOYED

n8n · Pinecone · Gemini Embeddings — 2026

Built and deployed a RAG-powered customer support agent for a real client (AJ Digital Agency, Mardan). Debugged the full vector store ingestion pipeline end-to-end using Pinecone and Gemini embeddings. Tuned system prompts specifically for lead generation conversion. Shipped to production.

NLP & DEEP LEARNING — FROM-SCRATCH IMPLEMENTATIONS

Transformer Self-Attention: Implemented the complete self-attention mechanism from first principles using NumPy — query/key/value projections, scaled dot-product attention, and multi-head concatenation. Built to understand the math, not import the abstraction.

Word2Vec Embeddings: Implemented Skip-gram and CBOW word embedding models from scratch. Trained on corpus data to produce dense vector representations capturing semantic relationships between words.

Text-to-Speech (TTS): Built a text-to-speech pipeline covering text normalization, phoneme mapping, and audio waveform synthesis. Implemented signal processing fundamentals for speech generation.

AutoInsight AI: Engineered an end-to-end automated ML pipeline — raw CSV input → EDA → model training → SHAP explainability → AI-generated plain-language writeup of results.

EDUCATION

BS Computer Science (Artificial Intelligence) Expected 2027

Abdul Wali Khan University, Mardan (Swat Campus)

CGPA: 3.51 / 4.00

Intermediate (FSc / ICS) 2021–2023

Govt. PG Jehanzeb College, Saidu Sharif, Swat

865 / 1100

CERTIFICATIONS

Deep Learning and Reinforcement Learning – Coursera

Machine Learning using Python – Coursera

Exploratory Data Analysis for Machine Learning – Coursera

Introduction to Databases – Coursera

Advanced C++ – SimpliLearn